



Sanghita Roy

Roll No.:23035010421

B.Sc. Hons. in Data Science and Artificial
Intelligence

Indian Institute Of Technology, Guwahati

+91-9382150804
r.sanghita@op.iitg.ac.in

Github | Website
linkedin.com/in/sanghita-roy-72b187332

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Sc. Hons.	Indian Institute of Technology, Guwahati	7.53 (Current)	2023-Present
Senior Secondary	WBCHSE Board	87.4%	2022
Secondary	WBBSE Board	93.4%	2020

EXPERIENCE

- IIT Guwahati**
Coordinator, ODP Welfare Club
 - Team Coordination
 - Event Planning & Execution
 - Speaker Outreach & Liaison
 - Community Engagement
 - Student Development Initiatives

March 2025 - Present
Guwahati, Assam

PROJECTS

- Movie Recommendation System**
Personal Project
 - Built a content-based movie recommendation system using cosine similarity on movie metadata from the TMDB dataset.
 - Performed data preprocessing and feature engineering in Python and integrated the model into a Flask-based web application.
 - Deployed the application on a cloud platform with OMDb API integration for real-time poster retrieval.
 - NHANES Age Group Classification**
Personal Project
 - Built a binary classifier to distinguish Adult (<65) vs Senior (>=65) using NHANES physiological and lifestyle data, with domain-driven feature engineering (HOMA-IR, glucose/insulin risk flags, BMI categories).
 - Implemented an end-to-end scikit-learn pipeline (imputation, scaling, encoding) with class-balanced Logistic Regression tuned via 5-fold Stratified CV using F1-score.
 - Evaluated model robustness using ROC-AUC, confusion matrix, and learning curves, and generated submission-ready predictions.
 - Customer Churn Prediction & Business Analytics Dashboard (Telco Customer Churn Dataset)**
Personal Project
 - Built a machine learning-based churn prediction system using the Telco Customer Churn dataset. Performed data cleaning (handling missing TotalCharges values), feature encoding, and feature alignment before applying a trained classification model to predict high-risk and low-risk customers based on optimized probability thresholds.
 - Developed a Flask-based web application that allows CSV upload and generates real-time churn predictions along with key business metrics including total customers, high-risk customers, revenue at risk, retention budget estimation, and potential profit calculation.
 - Integrated SHAP (SHapley Additive Explanations) to interpret model predictions and display top feature drivers influencing churn behavior. Designed an interactive dashboard using HTML, CSS, JavaScript, and Chart.js to visualize churn distribution, financial impact analysis, and feature importance in a business-friendly format.
 - NeuraScan AI: Explainable Brain Tumour Detection from MRI**
Personal Project
 - Developed an end-to-end deep learning system for multiclass brain tumour classification from MRI scans using transfer learning with VGG16, ResNet50, and EfficientNetB0. Performed image preprocessing, augmentation, and comparative evaluation using accuracy, precision, recall, F1-score, and ROC-AUC, achieving a best test accuracy of 90.4% with VGG16.
 - Designed a Flask-based web application that enables MRI image upload, real-time tumour prediction, Grad-CAM visual explainability, confidence estimation, automated PDF report generation, and AI-assisted result interpretation. Deployed the application on Hugging Face Spaces using Docker for cloud-based inference.
 - Implemented Grad-CAM to improve model interpretability by highlighting clinically relevant image regions and engineered a complete inference pipeline including image validation, preprocessing, prediction, explainability, and downloadable medical reports using TensorFlow, Keras, Python, Flask, and Hugging Face Spaces.

2025
2025
2026
2026

TECHNICAL SKILLS

- **Programming:** Python, R, MySQL, HTML, CSS
- **Libraries & Frameworks:** Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn, Flask
- **Tools:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code
- **Data Visualization Tools:** Power BI, Microsoft Excel

KEY COURSES TAKEN

- **Mathematics:** Linear Algebra, Basic Calculus, Probability & Statistics, Optimization
- **Core Courses:** Data Science, Data Analysis, Data Modeling & Visualization, Data Mining & Warehousing, Data Structures & Algorithms, AI Basics, RDBMS, Time Series Analysis & Forecasting, Machine Learning Fundamentals, Deep Learning Essentials

POSITIONS OF RESPONSIBILITY

- **Coordinator,** ODP Welfare Club, IIT Guwahati *March 2025 - Present*

ACHIEVEMENTS

- **Certificate of Appreciation,** Coordinator, Welfare Club, IIT Guwahati *2025*

EXTRA-CURRICULAR ACTIVITIES

Fine Arts (Painting)

Bangiya Sangeet Parishad (Ankan Kala Bibhag), affiliated to Rabindra Bharati University, Calcutta

- **Ankan Manjari** (Junior Diploma – 1st Year), 2015–2016
- **Ankan Sree** (Senior Diploma – 3rd Year), 2017–2018
- **Ankan Bivakar** (5th Year), 2019–2020
- Secured **First Division with Distinction** in both Theory and Practical examinations in all three levels.

Vocal Music

Bangiya Sangeet Parishad, affiliated to Rabindra Bharati University, Calcutta

- **Sangeet Manjari** (Junior Diploma – 1st Year), 2013–2014
- **Sangeet Sree** (Senior Diploma – 3rd Year), 2015–2016
- **Sangeet Bivakar** (5th Year), 2017–2018
- Secured **First Division with Distinction** in both Theory and Practical examinations in all three levels.

LANGUAGES

- Bengali – Native Proficiency
 - English – Working Proficiency
 - Hindi – Working Proficiency
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